ChatGPT

Drug Summary

Pembrolizumab (Keytruda) is a humanized IgG4 monoclonal antibody that targets the PD-1 checkpoint on T cells [1] [2]. By blocking PD-1, pembrolizumab prevents binding of its ligands (PD-L1/PD-L2) and thereby reactivates anti-tumor T-cell responses [1] [2]. In breast cancer, pembrolizumab has been approved primarily for triple-negative breast cancer (TNBC): as neoadjuvant therapy (with chemotherapy) followed by adjuvant use in high-risk early-stage TNBC, and as first-line therapy (with chemotherapy) in metastatic TNBC expressing PD-L1 [3] [4]. Approval requires PD-L1 expression (combined positive score ≥10) in metastatic TNBC [5]. It also has tissue-agnostic FDA approvals for MSI-H/dMMR tumors and for tumor mutational burden–high (≥10 mut/Mb) solid tumors, although these indications are rare in breast cancer [6] [7]. Common toxicities include immune-mediated inflammatory events (e.g. pneumonitis, colitis, endocrinopathies) and infusion reactions [8].

Identifiers & Synonyms

- **ChEMBL:** CHEMBL3137343 (Pembrolizumab)
- **DrugBank:** DB09037
- **BRAND:** Keytruda [9] (US)
- **Development codes:** MK-3475, SCH-900475 [9] (Merck codes)

Mechanism of Action (Breast Cancer Context)

Pembrolizumab specifically binds human PD-1 (encoded by *PDCD1*) on T lymphocytes and blocks its interaction with PD-L1/PD-L2 [1] [2]. PD-1 signaling normally inhibits T-cell activation and cytokine production to maintain immune tolerance [10]. By antagonizing PD-1, pembrolizumab "releases the brakes" on T cells, promoting cytotoxic T-cell infiltration and cytokine (e.g. IFN-γ) release within tumors [1] [10]. In TNBC (often an immunogenic subtype), this leads to enhanced anti-tumor immunity. Reactome pathway R-HSA-389948 ("Co-inhibition by PD-1") describes how active PD-1 signaling suppresses IL-2 and Bcl-xL to inhibit T-cell function [10]; pembrolizumab abrogates this inhibition, enabling T-cell expansion and tumor killing.

Primary Targets (Human)

- **PDCD1 (PD-1):** The sole direct target of pembrolizumab is the programmed cell death-1 receptor (PD-1) on T cells [1]. By blocking PD-1, pembrolizumab indirectly affects PD-L1/PD-L2–expressing tumor or immune cells.

Pathways Overview

Key pathways relevant to pembrolizumab's action in breast cancer include immune checkpoint and T-cell activation pathways. For example:

- **R-HSA-389948 (Co-inhibition by PD-1)** – An immune checkpoint pathway in Reactome. PD-1 engagement inhibits TCR signaling and IL-2 production, suppressing T cells [10]. Pembrolizumab blocks this pathway.
- **KEGG hsa05235 (PD-L1 expression and PD-1 checkpoint in cancer)** – Illustrates regulation of PD-L1/PD-1. Oncogenic MAPK, PI3K/Akt, STAT3 and NF-κB signaling drive PD-L1 upregulation; PD-L1 binding to PD-1 inactivates effector T cells [11].
- **R-HSA-202403 (T-cell receptor signaling)** – Activation of the T-cell receptor triggers cascades (including NF-κB) that lead to T-cell proliferation and effector function [12].
- **GO:0042110 (T cell activation)** – General process of activating T cells. Upstream of this are TCR and co-stimulatory signals (e.g. CD28).
- **HALLMARK_INTERFERON_GAMMA_RESPONSE** (MSigDB) – Genes induced by IFN-γ, representing antigen presentation and macrophage activation.
- **KEGG hsa04630 (Jak-STAT signaling pathway)** – Cytokine signaling including interferons. Engagement of IFN-γ receptors triggers JAK1/2, STAT1 phosphorylation, and transcription of immune-effector genes [13].
- **KEGG hsa04060 (Cytokine–cytokine receptor interaction)** – Broad array of cytokines (e.g. IL-2, IFNs, chemokines) and receptors governing immune responses. Cytokine signals coordinate innate and adaptive immunity [14].
- **R-HSA-170834 (Signaling by TGF-β receptor complex)** – TGF-β cytokines signal through SMADs; this pathway fosters immune tolerance and exclusion of cytotoxic T cells. Upregulation of TGF-β signaling is linked to immunotherapy resistance [15].
- **R-HSA-9758274 (Regulation of NF-κB signaling)** – NF-κB is a transcription factor family that drives inflammatory gene expression. It is activated by cytokines and by antigen (TCR) signals in T cells [12], promoting immune activation.
- **R-HSA-194138 (Signaling by VEGF)** – VEGF and its receptors promote angiogenesis (tumor blood vessel formation). VEGF-driven angiogenesis is associated with tumor growth and immune evasion [16].

Each pathway intersects with immune regulation. For instance, PD-1 blockade (Reactome R-HSA-389948) relieves suppression of TCR signaling [10]; meanwhile, higher activity of interferon and cytokine pathways (Jak-STAT, cytokine interactions) amplifies anti-tumor T-cell activity [13] [14]. Conversely, upregulation of TGF-β or VEGF pathways can counteract immune effects (see below).

Upregulated pathways (by context/subtype)

In responders, pembrolizumab is associated with increased activity of immune effector pathways. Neoadjuvant trials (e.g. KEYNOTE-522) showed higher rates of pathologic complete response, implying activation of immune and apoptotic programs [17]. Translational data suggest that IFN-γ response and cytotoxic T-cell gene signatures become upregulated after PD-1 blockade. For example, IFN-γ–induced chemokines (CXCL9/10) and perforin/granzyme expression rise in tumors responding to pembrolizumab. In contrast, immunosuppressive pathways (such as TGF-β signaling) are often relatively downregulated in responders [15]. Overall, pembrolizumab tends to upregulate T cell–related pathways (TCR signaling, IFN signaling) in sensitive tumors.

Downregulated pathways (by context/subtype)

Conversely, pembrolizumab treatment is linked to downregulation of pathways that inhibit immune activity. Notably, TGF-β signaling (R-HSA-170834) and related immune-exclusion programs are diminished in sensitive tumors [15]. Other inhibitory circuits – such as IL-10 production or alternative immune checkpoints (e.g. LAG3, TIM-3) – may likewise be suppressed in responders (although data are limited). Thus, downregulation of immune-evasive pathways accompanies pembrolizumab efficacy.

Sensitivity pathways (Upregulated)

Pathways whose upregulation predicts pembrolizumab sensitivity include classic immune-activation signatures. High tumor expression of PD-L1 (e.g. CPS≥10) is required for the approved indication [5], reflecting that PD-1 pathway activity must be present to target. Upregulated T-cell–inflamed pathways – including IFN-γ/JAK-STAT signaling, antigen processing, and TCR signaling – also correlate with response. Tumors with pre-existing interferon responses or high TIL levels (indicative of upregulated immune signaling) show better responses. For example, enhanced PD-1 pathway engagement (Reactome R-HSA-389948) in a tumor (i.e. high PD-L1/PD-1 interaction) paradoxically makes the cancer more vulnerable to pembrolizumab [10] [5]. (In other words, an “immune-hot” TME – marked by upregulated IFN and cytokine pathways – confers drug sensitivity.)

Sensitivity pathways (Downregulated)

Pathways that are downregulated in sensitive tumors tend to be immunosuppressive. For instance, lower activity of TGF-β or WNT/β-catenin pathways reduces immune exclusion [15]. Likewise, tumors with downregulated PD-L1 (in the sense of lower compensatory immunosuppression after therapy) may respond better. In practice, established sensitivity biomarkers – PD-L1 up and TGF-β down – reflect these patterns: high PD-1 pathway engagement (for drug binding) plus low TGF-β signaling yields optimal response [15] [5].

Resistance pathways (Upregulated)

Upregulation of certain pathways drives resistance to pembrolizumab. Chief among these is TGF-β signaling (R-HSA-170834): studies show high TGF-β pathway activation generates a T-cell–excluded, fibrotic tumor microenvironment that attenuates checkpoint blockade efficacy [15]. Other pathways implicated in resistance include WNT/β-catenin signaling and PI3K/MAPK activation (which can enhance PD-L1 expression but also promote immune evasion). Elevated VEGF signaling (R-HSA-194138) and angiogenesis likewise correlate with resistance by fostering immunosuppression [16]. Immune checkpoint cascades parallel to PD-1 (e.g. LAG3) may be co-upregulated, enabling escape. In TNBC, tumors that fail to respond often show upregulated inflammatory but inhibitory feedback loops (e.g. chronic NF-κB-driven inflammation without effective cytotoxicity).

Resistance pathways (Downregulated)

Resistance is also associated with downregulation of pro-immune pathways. Tumors that evade pembrolizumab often have impaired antigen presentation (e.g. reduced HLA class I), diminished T-cell receptor signaling, or loss of IFN-γ signaling components. For example, loss of PTEN (PI3K pathway) or defects in IFN-γ/JAK-STAT can blunt immunogenicity. While specific breast cancer data are sparse, general tumor-immune knowledge suggests that downregulation of antigen-processing genes and T-cell activation (GO:0042110, R-HSA-202403) can underlie checkpoint resistance.

Subtype-stratified findings

- **TNBC:** Pembrolizumab has clear efficacy. Early-stage high-risk TNBC achieves higher pathologic complete response and longer disease-free survival when pembrolizumab is added to neoadjuvant chemotherapy [18] [17]. In metastatic TNBC, KEYNOTE-355 showed that pembrolizumab + chemo improves PFS and OS in patients with PD-L1 CPS≥10 [19]. These effects are specific to TNBC; PD-L1 testing (CPS) is routine in metastatic TNBC to select therapy [5] [20].
- **HR+/HER2– BC:** Clinical benefit of pembrolizumab is not established. Current guidelines note **no definitive role** for PD-1 inhibitors in HR+/HER2– breast cancer, and the evidence (including small trials of pembrolizumab in endocrine-resistant disease) is insufficient [21]. Immune therapies are generally not recommended in this subtype outside trials.
- **HER2+ BC:** Similarly, there is no approved use of pembrolizumab for HER2-positive breast cancer, and trials combining PD-1 blockade with HER2-targeted therapy are ongoing. Routine use in HER2+ disease is not supported by current data [21].
- **gBRCA1/2-mutated BC:** BRCA-driven tumors often have higher mutational load, suggesting potential immunogenicity. However, pembrolizumab has no specific indication tied to BRCA status in breast cancer. Trials (e.g. KEYLYNK-001) are exploring pembrolizumab plus chemotherapy in HER2–/BRCA-mutant metastatic BC, but no results have yet altered practice. In summary, BRCA status is not presently an independent predictor of pembrolizumab response in breast cancer.

Contraindications and Safety

Pembrolizumab has **no absolute contraindications** listed in its FDA label [22]. The main safety concerns are immune-related adverse events: pneumonitis, colitis, hepatitis, nephritis, endocrinopathies (hypophysitis, thyroiditis), dermatologic reactions, and other inflammatory toxicities [8]. Infusion-related reactions can also occur. Special considerations include embryofetal risk: pembrolizumab can cause fetal harm, so women of childbearing potential must use effective contraception [23]. Monitoring for hepatic, renal, and endocrine function during therapy is recommended. Otherwise, the safety profile in breast cancer patients appears similar to that in other indications.

Trial and Guideline Context

Key clinical trials have demonstrated pembrolizumab’s benefit in TNBC. In the phase III KEYNOTE-522 trial (early-stage TNBC), adding pembrolizumab to neoadjuvant chemotherapy significantly improved pathological complete response rates and event-free survival [17]. In the phase III KEYNOTE-355 trial (metastatic TNBC), pembrolizumab + chemotherapy significantly prolonged PFS and OS versus chemotherapy alone in PD-L1–positive (CPS≥10) patients [24]. Based on these data, major guidelines now endorse pembrolizumab for TNBC: e.g. NICE TA851 (Dec 2022) recommends neoadjuvant + adjuvant pembrolizumab in high-risk early TNBC [18], and NICE TA801 (Jun 2022) recommends pembrolizumab + chemo in untreated PD-L1–positive mTNBC [19]. The 2024 Chinese expert consensus similarly states that PD-1 inhibitors should be used in early-stage II–III and first-line metastatic TNBC [21]. In contrast, guidelines agree that there is **insufficient evidence** to support pembrolizumab in HR+/HER2– or HER2+ breast cancer [21]. ASCO Rapid Guidelines (2022) also updated early TNBC treatment to include pembrolizumab with chemotherapy in the perioperative setting, reflecting these trial results.

Additional mechanistic/clinical notes

- **Biomarker testing:** PD-L1 testing (CPS) is required for metastatic TNBC. Pembrolizumab is indicated only if CPS≥10 [5]. Assays such as the FDA-approved PD-L1 IHC (22C3 pharmDx) are used.
- **TMB/MSI status:** While rare in breast cancer, tumors with high microsatellite instability (MSI-H/dMMR) or high tumor mutational burden (TMB ≥10 mut/Mb) are approved pembrolizumab indications across cancers [6] [7]. Testing for MSI/TMB may be considered in refractory breast cancers to identify candidates.
- **Predictive signatures:** Research suggests that an “immune-inflamed” gene signature (high IFN-γ, T-effector genes) predicts benefit, whereas an “immune-desert” or myeloid-rich signature predicts resistance. Clinical trials are exploring composite biomarkers (e.g. 27-gene IO score) to refine patient selection in breast cancer.
- **Combination regimens:** Ongoing studies combine pembrolizumab with targeted therapies (e.g. PARP inhibitors, CDK4/6 inhibitors, ADCs) in various breast cancer subtypes. Early results indicate potential synergy, but these approaches remain experimental.
- **Safety guidelines:** Management of immune-related AEs follows oncology guidelines (e.g. corticosteroids for grade ≥2 irAEs). No breast-specific contraindications are noted beyond general ICI cautions.

Pathway Evidence Table

Pathway (ID/ Name)	Regulation	Effect of Pembrolizumab (↑ = Sensitive/ ↓ = Resistant)	Biological Rationale (breast cancer context)	Reference(s)
Reactome R-HSA-389948 – **PD-1 co-inhibition**	Up	Resistant	Active PD-1 signaling inhibits TCR activation and IL-2 production, suppressing T cells [10]. Pembrolizumab blocks this, so tumors with active PD-1 are sensitive.	[10]
KEGG hsa05235 – **PD-L1/PD-1 checkpoint in cancer**	Up	Sensitive	High tumor PD-L1 (driven by MAPK/STAT3 pathways) engages PD-1 to evade immunity [11]; pembrolizumab exploits this by binding PD-1, so PD-L1↑ tumors respond.	[11]

Pathway (ID/ Name)	Regulation	Effect of Pembrolizumab (↑ = Sensitive/ ↓ = Resistant)	Biological Rationale (breast cancer context)	Reference(s)
Reactome R-HSA-202403 – **TCR signaling**	Up	Sensitive	Upregulated TCR signaling leads to NF-κB activation and T-cell proliferation [12] , promoting an effective immune response. Pembrolizumab augments this by unleashing T cells.	[12]
GO:0042110 – **T cell activation**	Up	Sensitive	Enhanced T-cell activation (via co-stimulation and cytokines) drives tumor cell killing. Upregulated activation pathways indicate an "inflamed" tumor microenvironment favorable to PD-1 blockade.	General knowledge
KEGG hsa04630 – **Jak-STAT signaling**	Up	Sensitive	Cytokines (e.g. IFN-γ, IL-2) signal via JAK/STAT to induce anti-tumor genes [13] . High Jak-STAT activity (e.g. from IFN signaling) correlates with pembrolizumab efficacy.	[13]
KEGG hsa04060 – **Cytokine–cytokine receptor**	Up	Sensitive	Broad cytokine signaling (IFNs, ILs, chemokines) regulates innate/adaptive immunity [14] . Elevated cytokine pathway expression indicates active immune recruitment, enhancing PD-1 inhibitor response.	[14]
Reactome R-HSA-170834 – **TGF-β signaling**	Up	Resistant	TGF-β pathway drives immune suppression and exclusion of CD8+ T cells [15] . Tumors with upregulated TGF-β signaling typically evade immune attack, reducing pembrolizumab sensitivity.	[15]

Pathway (ID/ Name)	Regulation	Effect of Pembrolizumab (↑ = Sensitive/ ↓ = Resistant)	Biological Rationale (breast cancer context)	Reference(s)
Reactome R-HSA-9758274 – **NF-κB signaling**	Up	Sensitive	NF-κB controls pro-inflammatory and survival genes in immune cells [12]. Enhanced NF-κB activity in T cells promotes cytokine production and T-cell proliferation, aiding checkpoint blockade efficacy.	[12]
Reactome R-HSA-194138 – **VEGF signaling**	Up	Resistant	VEGF drives angiogenesis and vascular permeability [16], which can create an immunosuppressive TME and inhibit T-cell infiltration. High VEGF pathway activity is linked to reduced response to immunotherapy.	[16]

Each pathway above is supported by published pathway analyses or clinical evidence. For example, Reactome's PD-1 co-inhibition pathway (R-HSA-389948) explicitly shows how PD-1 engagement inhibits T-cell functions [10], explaining why pembrolizumab's blockade of this pathway produces a sensitive phenotype. Conversely, KEGG's VEGF signaling pathway shows VEGF-driven tumor angiogenesis [16], rationalizing why its upregulation confers resistance to T-cell–mediated therapies. These entries integrate the mechanistic basis (pathway biology) with observed pembrolizumab response patterns in breast cancer, as supported by the cited sources.

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